

Human Inspired Genetic Algorithms for Multi-Agent Evacuation Routing

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Declaration

This dissertation is submitted to the University of Bath in accordance with the requirements of the degree of Master of Science in the Department of Computer Science. No portion of the work in this dissertation has been submitted in support of an application for any other degree or qualification of this or any other university or institution of learning. Except where specifically acknowledged, it is the work of the author.

Abstract

⟨The abstract should appear here. An abstract is a short paragraph describing the aims of the project, what was achieved and what contributions it has made.⟩

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Acknowledgements

Add any acknowledgements here.

This research has been conducted in line with the University of Bath's ethics policy and ethical approval has been obtained through Ethics@bath: <insert ethics@bath project ID here, separate by commas if you have made more than one ethics submission>

Chapter 1

Introduction

1.1 Background

1.1.1 Evacuation Routing Overview

... and subsections ...

Example Sub-subsection

... and sub-subsections.

1.2 Another Section With a Long Title and Whose Title Is Abbreviated in the Table of Contents

Table 1.1: An example table

Items	Values
Item 1	Value 1
Item 2	Value 2

Another section, just for good measure. You can reference a table, figure or equation using `\ref`, just like this reference to Table 1.1.

1.3 Example Lists

1.3.1 Enumerated

1. Example enumerated list:
 - a nested enumerated list item;
 - and another one.
2. Second item in the list.

1.3.2 Itemised

- Example itemised list.
 - A nested itemised list item.
- Second item in the list.

1.3.3 Description

Item 1 First item in the list.

Item 2 Second item in the list.

Chapter 2

Literature and Technology Survey

This is the chapter for your Literature and Technology Survey.

You will wish to cite authors like (Brünnler, 2003) or (Straßburger, 2017). Alternate commands are used to cite Brünnler (2003) as a noun, or add text to the citation, (e.g., Straßburger, 2017). You can also use other styles of citation, for example numerical. Whatever style you use, be consistent.

Chapter 3

Requirements

If you are doing a primarily software development project, this is the chapter in which you review the requirements decisions and critique the requirements process.

Chapter 4

Design

This is the chapter in which you review your design decisions at various levels and critique the design process.

Chapter 5

Implementation and Testing

This is the chapter in which you review the implementation and testing decisions and issues, and critique these processes.

Code can be output inline using `\lstinline|some code|`. For example, this code is inline: `public static int example = 0;` (we have used the character `|` as a delimiter, but any non-reserved character not in the code text can be used.)

Code snippets can be output using the `\begin{lstlisting} ... \end{lstlisting}` environment with the code given in the environment. For example, consider listing 5.1, below.

Listing 5.1: Example code

```
public static void main() {  
  
    System.out.println("Hello World");  
  
}
```

Code listings are produced using the package 'listings'. This has many useful options, so have a look at the package documentation for further ideas.

Chapter 6

Results

This is the chapter in which you present the results. You may include user evaluation here too.

Chapter 7

Discussion

This is the chapter in which you clearly review the results, and critique the process. You must end this chapter with a discussion of the implication of the results, acknowledging the limitations of your work, and identifying possible future work.

Chapter 8

Conclusions

This is the chapter in which you summarise the entire project, review the major achievements in the light of your original objectives.

Number of words until this point, excluding front matter: XXX.

References

- Brünnler, K., 2003. Atomic cut elimination for classical logic [Online]. In: M. Baaz and J.A. Makowsky, eds. *Computer science logic (csl)*. Springer, *Lecture Notes in Computer Science*, vol. 2803, pp.86–97. Available from: https://doi.org/10.1007/978-3-540-45220-1_9.
- Straßburger, L., 2017. Combinatorial flows and their normalisation [Online]. In: D. Miller, ed. *2nd international conference on formal structures for computation and deduction (fscd)*. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, *Leibniz International Proceedings in Informatics (LIPIcs)*, vol. 84, pp.31:1–17. Available from: <https://doi.org/10.4230/LIPIcs.FSCD.2017.31>.

Appendix A

Design Diagrams

Appendix B

User Documentation

Appendix C

Raw Results Output

Appendix D

Code

D.1 File: yourCodeFile.java

```
// This is an example java code file , just for illustration  
purposes  
public static void main() {
```

```
    System.out.print ("Hello World");  
}
```